

# PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes: Introduction & Continuous Wavelet Transform (CWT)

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## 1. Introduction and Historical Development

### Definition of a Wavelet

A **wavelet** is a localized, oscillatory wave-like function of effectively finite duration, with a zero mean. Unlike a sine wave, which extends infinitely in both directions, a wavelet grows and decays over a short period, providing simultaneous **time** and **frequency** localization.

### 1.1 Historical Origin of Wavelets

The development of wavelet theory is a synthesis of ideas spanning mathematics, physics, and digital signal processing:

- **1910 (Alfréd Haar):** Proposed the first and simplest wavelet, the **Haar wavelet**, as an orthonormal basis for functions. It consists of a step function that changes value abruptly between  $+1$  and  $-1$ .
- **1946 (Dennis Gabor):** Developed the Windowed Fourier Transform (or Gabor Transform) to introduce time localization using Gaussian window envelopes.
- **1981 (Jean Morlet):** A geophysicist processing seismic data introduced the concept of the “wavelet of constant shape” with variable scaling and translation to achieve depth resolution.
- **1984 (Alex Grossmann):** Collaborated with Morlet to establish the rigorous mathematical framework for the Continuous Wavelet Transform (CWT) using group theory.
- **1986 (Stéphane Mallat & Yves Meyer):** Formulated **Multi-resolution Analysis (MRA)**, establishing a unified foundation connecting wavelets to subband digital filtering.
- **1988 (Ingrid Daubechies):** Constructed the first family of continuous, compactly supported, orthonormal wavelet bases with maximum vanishing moments (the **Daubechies wavelets**), making wavelets highly practical for computer implementation.

### 1.2 Different Communities of Wavelet

Wavelet theory is uniquely interdisciplinary, developed independently across multiple scientific domains:

1. **Mathematics Community:** Studied under *Calderón-Zygmund operators*, singular integrals, and harmonic analysis. Focuses on basis representations of Hilbert and Banach function spaces ( $L^2(\mathbb{R})$ ).
2. **Geophysics & Physics Community:** Morlet and Grossmann used wavelets for seismic reflection parsing and quantum mechanics (coherent states, path integrals).
3. **Digital Signal Processing (DSP) & Engineering Community:** Developed wavelets as **filter banks** (subband coding) for audio, image, and video compression.

## 2. Time-Frequency Analysis Limitations

Traditional signal analysis relies on the **Fourier Transform (FT)**, which maps a time-domain signal  $x(t)$  to its frequency spectrum  $X(f)$ :

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

- **Limitation:** FT has *infinite time support* ( $e^{-j2\pi ft}$  spans from  $-\infty$  to  $\infty$ ). It yields perfect frequency resolution but completely destroys all time information (we know *what* frequencies exist, but not *when* they occur).

## 2.1 Short-Time Fourier Transform (STFT)

To introduce time localization, the **Short-Time Fourier Transform (STFT)** multiplies the signal with a sliding window function  $w(t)$  centered at time  $\tau$ :

$$STFT_x(\tau, f) = \int_{-\infty}^{\infty} x(t)w^*(t - \tau)e^{-j2\pi ft} dt$$

- **Limitation:** The window width is **fixed** for all frequencies.
- **Heisenberg's Uncertainty Principle:** The resolutions in time ( $\Delta t$ ) and frequency ( $\Delta f$ ) are bounded:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

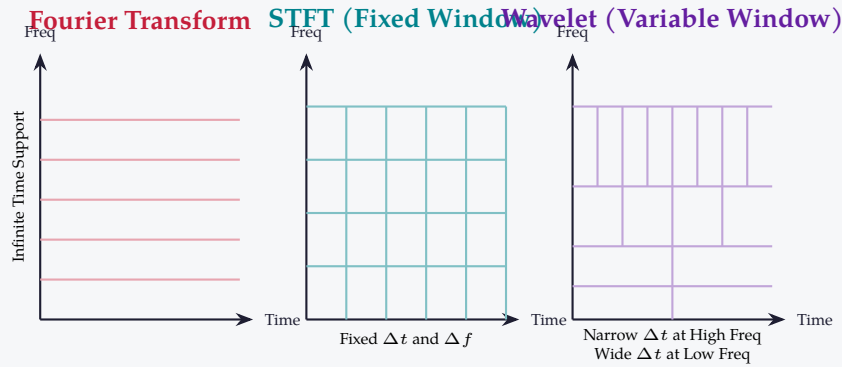
Under STFT, we must choose a fixed window. A narrow window gives good time resolution but poor frequency resolution (Wideband STFT). A wide window gives good frequency resolution but poor time resolution (Narrowband STFT).

## 2.2 The Wavelet Multi-resolution Approach

The Continuous Wavelet Transform (CWT) resolves this restriction by using a **variable-width window**:

- At **high frequencies**, it uses narrow windows (high time resolution, lower frequency resolution) to capture rapid transients and edges.
- At **low frequencies**, it uses wide windows (high frequency resolution, lower time resolution) to capture slow-varying stationary components.

## Time-Frequency Tiling Grids Comparison



### 3. Continuous Wavelet Transform (CWT)

#### Definition of CWT

The **Continuous Wavelet Transform (CWT)** of a continuous-time signal  $x(t) \in L^2(\mathbb{R})$  is the correlation of the signal with a scaled and translated version of a mother wavelet function  $\psi(t)$ :

$$CWT_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left( \frac{t-b}{a} \right) dt$$

where:

- $a \in \mathbb{R} \setminus \{0\}$  is the **scale parameter** (controls compression/dilation).
- $b \in \mathbb{R}$  is the **translation parameter** (controls shift along the time axis).
- $\psi(t)$  is the **mother wavelet**, and  $\psi^*(t)$  denotes its complex conjugate.
- $\frac{1}{\sqrt{|a|}}$  is the normalization factor ensuring energy is invariant across scales.

#### 3.1 Scaling and Translation Parameters

- **Scaling ( $a$ ):** Scaling either dilates (stretches) or compresses the mother wavelet:
  - **Large scale** ( $|a| > 1$ ): Wavelet is dilated, representing a low-frequency, long-duration window (captures coarse features).
  - **Small scale** ( $|a| < 1$ ): Wavelet is compressed, representing a high-frequency, short-duration window (captures fine details and transients).
- **Translation ( $b$ ):** Moves the wavelet along the time axis. It indicates the time center of the window.

#### 3.2 The Constant Q-Factor Filtering Interpretation

The ratio of the center frequency  $f_c$  of a filter to its bandwidth  $\Delta f$  is defined as the **Quality Factor (Q-factor)**:

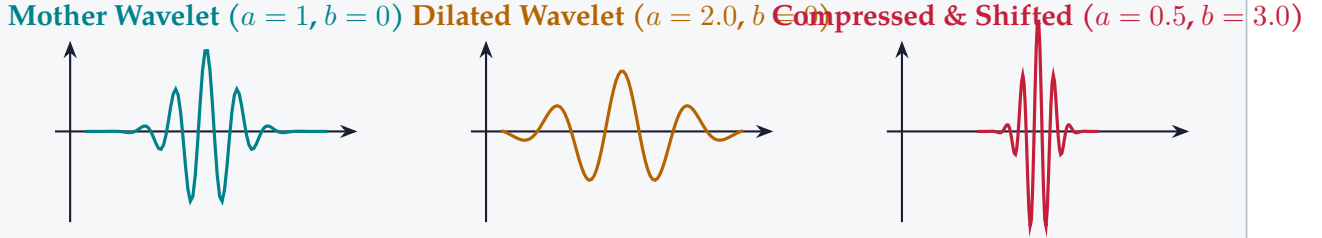
$$Q = \frac{f_c}{\Delta f}$$

In CWT, compressing the wavelet in time by  $a$  stretches its Fourier spectrum by  $1/a$ . Thus, center frequency and bandwidth scale proportionally:

$$f_c(a) = \frac{f_c(1)}{a}, \quad \Delta f(a) = \frac{\Delta f(1)}{a} \implies Q = \frac{f_c(a)}{\Delta f(a)} = \text{Constant}$$

Hence, the CWT behaves as a **constant-Q bandpass filter bank**.

#### Wavelet Dilation and Compression



## 4. Mathematical Properties of CWT

### 4.1 CWT as a Linear Operator

The CWT is a **linear operator**. If  $x(t) = c_1 x_1(t) + c_2 x_2(t)$ , then:

$$CWT_x(a, b) = c_1 CWT_{x_1}(a, b) + c_2 CWT_{x_2}(a, b)$$

### 4.2 Admissibility Condition

For a function  $\psi(t)$  to be a valid mother wavelet, it must satisfy the **admissibility condition**:

$$C_\psi = \int_{-\infty}^{\infty} \frac{|\Psi(f)|^2}{|f|} df < \infty$$

where  $\Psi(f)$  is the Fourier transform of  $\psi(t)$ .

- **Consequence:** For the integral to be finite,  $\Psi(0) = 0$ . This implies:

$$\int_{-\infty}^{\infty} \psi(t) dt = 0$$

The average value (or DC component) of a wavelet must be **zero**. It must oscillate above and below the zero axis.

### 4.3 Inverse Continuous Wavelet Transform

If the admissibility condition  $C_\psi < \infty$  is met, the signal  $x(t)$  can be reconstructed from its wavelet coefficients  $CWT_x(a, b)$  via the inverse CWT formula:

$$x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right) \frac{da db}{a^2}$$

#### 4.4 Energy Conservation (Parseval's Relation)

The wavelet transform preserves energy. If  $x(t) \in L^2(\mathbb{R})$ , then:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |CWT_x(a, b)|^2 \frac{da db}{a^2}$$

## Quick Revision — Module 1

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### Key Wavelet Formulas and Metrics

|                                     |                                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Continuous Wavelet Transform</b> | $CWT_x(a, b) = \frac{1}{\sqrt{ a }} \int_{-\infty}^{\infty} x(t) \psi^* \left( \frac{t-b}{a} \right) dt$                                                       |
| <b>Admissibility Condition</b>      | $C_\psi = \int_{-\infty}^{\infty} \frac{ \Psi(f) ^2}{ f } df < \infty \implies \int_{-\infty}^{\infty} \psi(t) dt = 0$                                         |
| <b>Inverse Wavelet Transform</b>    | $x(t) = \frac{1}{C_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} CWT_x(a, b) \frac{1}{\sqrt{ a }} \psi \left( \frac{t-b}{a} \right) \frac{da db}{a^2}$ |
| <b>Heisenberg Uncertainty</b>       | $\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \quad (\text{Resolution limit in Time-Frequency plane})$                                                          |
| <b>Constant Q-Factor</b>            | $Q = \frac{f_c}{\Delta f} = \text{Constant} \quad (\text{Proportional frequency/bandwidth scaling})$                                                           |

### One-Line Definitions

- **Wavelet:** A localized, oscillatory function of finite duration with a zero mean value.
- **CWT:** The correlation of a continuous-time signal with a continuously scaled and translated mother wavelet.
- **Scale Parameter ( $a$ ):** Controls the compression (small  $a$ , high frequency) or dilation (large  $a$ , low frequency) of a wavelet.
- **Translation Parameter ( $b$ ):** Determines the temporal location of the wavelet window center along the time axis.
- **Admissibility Condition:** A mathematical constraint requiring a wavelet's DC component to be zero for reconstruction to be possible.
- **Fourier Transform:** Spectral mapping with infinite time support, providing no temporal location details.
- **STFT:** Time-frequency mapping that uses a fixed window width, imposing rigid time-frequency resolutions.

### Frequently Asked / Viva Questions

#### Q1. Why can the Fourier Transform not analyze non-stationary signals?

The Fourier Transform integrates over all time ( $-\infty$  to  $\infty$ ), meaning it assumes the signal's spectral components do not change over time. It has no time localization, so it cannot identify \*when\* a transient event or change in frequency occurs.

#### Q2. How does the Short-Time Fourier Transform (STFT) attempt to fix the limitations of the Fourier Transform?

**Frequently Asked / Viva Questions (contd.)**

The STFT divides the signal into short segments by multiplying it with a sliding window function. The Fourier Transform is then computed for each segment, providing a time-frequency spectrogram.

**Q3. What is the drawback of the STFT?**

The STFT uses a fixed window size for all frequencies. Due to Heisenberg's Uncertainty Principle ( $\Delta t \cdot \Delta f \geq 1/4\pi$ ), a narrow window gives good time resolution but poor frequency resolution, while a wide window does the opposite.

**Q4. How do Wavelets resolve the resolution trade-off of the STFT?**

Wavelets use variable-width windows: narrow windows (small scales) at high frequencies for capturing rapid transitions, and wide windows (large scales) at low frequencies for capturing steady long-term behaviors.

**Q5. State the physical meaning of the scale parameter 'a'.**

Scale 'a' is inversely proportional to frequency. A small scale ( $a < 1$ ) compresses the wavelet, corresponding to high-frequency transients. A large scale ( $a > 1$ ) dilates the wavelet, corresponding to slow-varying low-frequency components.

**Q6. What is the significance of the admissibility condition?**

The admissibility condition ensures that the wavelet does not carry a DC component ( $\int_{-\infty}^{\infty} \psi(t)dt = 0$ ). This guarantees that the inverse wavelet transform exists and enables perfect reconstruction of the signal.

**Q7. Explain the constant Q-factor filtering property of the CWT.**

In CWT, as scale  $a$  changes, the center frequency and bandwidth of the wavelet filter change proportionally. Since  $Q = f_c/\Delta f$ , the quality factor remains constant across all scales, behaving as a logarithmic bandpass filter bank.

**Q8. What is the difference between Gabor transform and Wavelet transform?**

The Gabor transform is a special case of STFT using a Gaussian window. It uses a fixed window envelope across all frequencies. The Wavelet transform uses a scaled envelope that compresses at higher frequencies and expands at lower frequencies.

**Q9. Is the wavelet transform a redundant representation? Why?**

Yes. In the Continuous Wavelet Transform, the parameters 'a' and 'b' vary continuously, mapping a 1D time signal  $x(t)$  to a 2D time-scale plane  $CWT(a, b)$ . This generates highly redundant information, which is resolved by discretizing scales and shifts.

## Summary Comparison: FT, STFT, and CWT

| Feature         | Fourier Transform (FT)           | Short-Time FT (STFT)                | Continuous Wavelet (CWT)                  |
|-----------------|----------------------------------|-------------------------------------|-------------------------------------------|
| Time Resolution | None (integrates over all time). | Fixed (determined by window width). | Variable (excellent at high frequencies). |
| Freq Resolution | Perfect (at all frequencies).    | Fixed (determined by window width). | Variable (excellent at low frequencies).  |
| Window Shape    | Infinite ( $e^{-j2\pi ft}$ ).    | Fixed envelope ( $w(t)$ ).          | Scaled mother wavelet ( $\psi(t/a)$ ).    |
| Best Use Case   | Stationary signals.              | Quasi-stationary signals.           | Non-stationary/transient signals.         |